

PORT-MENIER, QC, Canada

WMO: 718100

Lat: 49.839N

Lon: 64.293W

Elev: 181

StdP: 14.60

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.9	(c) -4.7	(d) -16.5	(e) 2.3	(f) -7.3	(g) -12.4	(h) 2.8	(i) -3.9	(j) 33.4	(k) 21.3	(l) 29.8	(m) 17.9	(n) 6.1	(o) 290	(p) 0.722

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	74.8	64.2	72.0	62.7	69.7	61.2	66.5	71.8	64.7	69.3	63.1	67.5	10.5	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	91.5	68.4	62.8	86.1	67.1	61.2	81.4	65.6	31.3	71.8	30.0	69.4	28.8	67.4	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 26.6	(b) 23.1	(c) 20.3	(e) -17.4	(f) 81.1	(g) 4.0	(h) 3.2	(i) -20.2	(j) 83.4	(k) -22.6	(l) 85.3	(m) -24.8	(n) 87.0	(o) -27.7	(p) 89.4		
(4) 26.6	(5) 23.1	(6) 20.3	(7) -17.6	(8) 70.2	(9) 3.9	(10) 2.0	(11) -20.4	(12) 71.6	(13) -22.7	(14) 72.8	(15) -25.0	(16) 73.9	(17) -27.8	(18) 75.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.2	12.9	13.0	21.1	32.3	43.5	54.3	61.6	60.8	52.4	41.5	30.8	20.3		
	DBStd	18.54	9.80	9.92	9.24	6.18	5.55	5.78	4.27	4.35	5.72	6.17	7.60	8.81		
	HDD50	5653	1149	1035	894	530	215	23	0	0	38	271	576	922		
	HDD65	10183	1614	1455	1359	980	667	324	121	143	379	728	1026	1387		
	CDD50	977	0	0	0	0	13	151	361	334	110	8	0	0		
	CDD65	33	0	0	0	0	0	3	17	12	1	0	0	0		
	CDH74	113	0	0	0	0	0	16	61	33	3	0	0	0		
	CDH80	6	0	0	0	0	0	2	3	1	0	0	0	0		
	WSAvg	9.0	9.5	9.3	9.6	9.6	8.6	8.6	8.4	8.1	8.6	8.9	9.0	9.4		
	PrecAvg	48.0	4.3	3.7	3.5	3.1	3.6	3.7	4.1	4.2	4.1	4.8	4.4	4.6		
(15) Precipitation	PrecMax	57.2	6.1	6.3	8.3	5.5	7.4	6.7	6.4	7.6	7.5	8.7	6.9	11.1		
	PrecMin	37.3	1.6	1.4	0.9	1.0	1.0	1.7	2.1	0.6	1.1	1.6	1.9	1.9		
	PrecStd	5.1	1.3	1.1	1.7	1.3	1.6	1.1	1.2	1.4	1.5	1.6	1.3	1.8		
	PrecStd	5.1	1.3	1.1	1.7	1.3	1.6	1.1	1.2	1.4	1.5	1.6	1.3	1.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.7	37.3	42.5	52.7	67.6	75.8	79.1	77.9	71.3	60.5	50.9	41.6		
		MCWB	36.2	36.2	36.6	42.7	53.9	62.6	66.7	66.0	61.6	53.8	49.8	41.1		
	2%	DB	34.2	34.0	38.1	47.2	61.8	71.0	75.5	74.1	67.2	56.3	46.8	38.2		
		MCWB	33.5	32.9	34.2	40.0	51.7	59.8	64.5	64.1	59.5	51.9	45.4	37.3		
	5%	DB	31.6	31.6	35.6	43.7	57.4	67.8	72.6	71.4	64.2	53.3	43.8	35.4		
		MCWB	30.9	30.6	32.4	38.3	48.2	57.4	63.3	62.7	58.1	50.3	42.2	34.3		
	10%	DB	27.9	28.2	33.4	40.8	53.8	64.7	70.2	69.1	61.4	50.9	41.1	33.0		
		MCWB	27.0	26.8	31.0	36.8	46.6	56.0	62.0	61.4	56.3	47.8	39.6	31.7		
	0.4%	WB	36.7	36.8	38.0	44.5	57.1	66.2	69.8	68.5	64.6	55.8	50.2	41.1		
		MCDB	37.1	37.2	41.0	48.0	64.1	73.8	76.1	74.3	68.0	58.5	50.4	41.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.6	33.1	35.1	41.7	52.4	62.0	66.9	66.2	61.7	53.0	45.9	37.4		
		MCDB	34.2	33.6	37.0	45.5	58.9	67.5	72.2	71.5	65.2	55.2	46.6	37.8		
	5%	WB	31.0	30.6	33.1	39.4	49.9	59.6	65.2	64.6	59.3	50.8	42.5	34.3		
		MCDB	31.6	31.5	34.9	42.7	56.0	65.5	69.9	69.0	62.5	53.1	43.4	35.1		
	10%	WB	27.1	26.9	31.4	37.2	47.4	57.3	63.6	63.0	57.3	48.2	39.8	31.7		
		MCDB	27.9	28.0	33.3	40.2	52.7	62.7	68.2	67.1	60.6	50.2	40.9	32.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.8	14.2	14.2	12.3	14.8	15.5	14.8	14.3	13.7	11.5	10.1	11.4		
		MCDBR	15.8	11.9	13.6	16.8	22.1	20.4	18.5	17.4	17.0	14.8	11.6	10.7		
		MCWBR	16.0	11.7	10.8	11.1	13.2	10.7	9.6	9.4	11.0	11.5	11.1	11.1		
	5% WB	MCDBR	16.0	11.6	13.0	14.5	19.8	17.2	15.1	14.6	14.7	13.1	11.2	11.3		
		MCWBR	16.3	11.6	11.4	10.7	12.7	10.2	9.1	9.0	10.9	11.3	11.3	11.8		
			(d) 13.8	(e) 14.2	(f) 14.2	(g) 12.3	(h) 14.8	(i) 15.5	(j) 14.8	(k) 14.3	(l) 13.7	(m) 11.5	(n) 10.1	(o) 11.4		
(40) Clear-Sky Solar Irradiance	taub		0.271	0.276	0.298	0.338	0.357	0.376	0.410	0.377	0.352	0.333	0.295	0.275		
	taud		2.411	2.346	2.377	2.385	2.419	2.388	2.356	2.453	2.483	2.505	2.512	2.413		
	Ebn at Noon		251	278	291	288	286	280	268	273	268	254	241	231		
	Edh at Noon		20	27	31	35	35	36	37	32	28	23	18	17		
			(d) 0.271	(e) 0.276	(f) 0.298	(g) 0.338	(h) 0.357	(i) 0.376	(j) 0.410	(k) 0.377	(l) 0.352	(m) 0.333	(n) 0.295	(o) 0.275		
(44) All-Sky Solar Radiation	RadAvg		312	624	1035	1412	1649	1798	1712	1470	1104	674	342	238		
	RadStd		33	50	89	115	100	156	156	76	76	48	35	24		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	+2.23	N/A	+0.59	+0.65	N/A	N/A	N/A	N/A

Nomenclature: See separate page